

STAGE 1 PROBLEM IDENTIFICATION

Project Title

Designing a Student Record Management Database for Mtoko Government High School

Problem Description

At Mtoko Government High School, student records such as personal details, subject choices, and academic results are currently stored manually in physical files. This system is inefficient, time-consuming, and prone to loss or misplacement of important data. Retrieving student information takes a lot of time and updating records is difficult.

Statement of Intent

The aim of this project is to design and implement a computerized database system that allows the school to store, update, and retrieve student records quickly and accurately. The system will simplify data management and improve record security and accessibility.

Design Specification

- *Software to Use:* Microsoft Access (or any other simple DBMS software)

- *Key Tables:*

- Students (ID, Name, Age, Gender, Class, Admission Number)

- Subjects (Subject Code, Subject Name)

- Results (Student ID, Subject Code, Term, Marks)

- *Features:*

- Data entry forms for each table

[07/09, 19:59] ChatGPT: - Queries to search for students by name, class, or ID

- Reports for generating mark sheets and summaries

- User-friendly interface for teachers or admin staff

- *Security:* Password protection or user login

- *Backup:* Option to export or save a backup copy

STAGE 2 INVESTIGATION OF RELATED IDEAS

EXISTING SOLUTIONS

1. Manual Filing System (Current Method

Advantages:

- Simple and low-cost
- Requires no electricity or computer skills
- Easy to use in low-resource environments

Disadvantages:

- Time-consuming to update and search
- Prone to data loss, damage, or misplacement
- Difficult to generate reports or analyze data

2. Spreadsheet Software (e.g., Microsoft Excel)

Advantages:

- Easy to learn and use
- Supports sorting, filtering, and basic calculations
- Readily available on most computers

Disadvantages:

- Not secure for sensitive data
- Difficult to manage large datasets
- No relational structure (poor for linking multiple tables)

3. Cloud-Based Systems (e.g., Google Sheets or Online Platforms)

Advantages:

- Accessible from any device with internet
- Supports collaboration among staff

- Automatically saved and backed up

Disadvantages:

- Requires stable internet connection
- Data privacy concerns
- Some features may require paid subscriptions

4. Database Management System (DBMS) like Microsoft Access(Chosen Solution

Advantages:

- Handles large amounts of structured data
- Supports queries, reports, and user forms
- Can be secured with passwords and restricted access

Disadvantages:

- Requires basic training to use effectively
- Needs a computer with sufficient memory/storage
- Not easily accessible remotely without setup

each merged solution:

STAGE 3 GENERATION OF IDEAS

1. Microsoft Access + Manual Filing System

Explanation:

In this approach, Microsoft Access is used to manage digital records, while the manual filing system is kept as a physical backup for essential documents such as admission forms or exam reports.

Strengths:

- Ensures data is available even during power outages
- Manual files can act as legal or hard-copy references
- Provides an easy backup method in case of technical failure

Weaknesses:

- Requires double entry of some data (manual and digital)
- Increases workload and storage space
- Manual files are still vulnerable to loss or damage

2. Microsoft Access + Spreadsheet Software (Excel)

Explanation:

Data is first collected or stored in spreadsheets (e.g., class lists, test marks), then imported into the Microsoft Access database for structured storage and reporting.

Strengths:

- Spreadsheets are easy and familiar for quick data entry
- Acts as a temporary data collection tool when Access is not available
- Data can be imported/exported easily

Weaknesses:

- Data duplication may cause inconsistencies if not updated properly
- Spreadsheets don't have the same security or validation as databases
- Formatting differences can lead to import errors

3. Microsoft Access + Cloud-Based Backup (e.g., Google Drive)

Explanation:

The database file created in Microsoft Access is regularly saved to cloud storage to ensure safe, remote backup and easy restoration if needed.

Strengths:

- Data is safe even if the local system crashes
- Enables access from other computers when needed
- Automated backup can reduce risk of loss

Weaknesses:

- Internet access is required to upload/download files
- Risk of unauthorized access if cloud account is not secured
- May require basic training on cloud platforms

STAGE 4 DEVELOPMENT OF IDEAS

Selected Idea

Microsoft Access + Cloud-Based Backup

Justification for Selected Idea

1. Structured Data Management

Microsoft Access allows for the creation of multiple related tables, queries, and reports, making it ideal for managing large sets of student data.

2. Improved Data Security

With password protection and access controls, Access provides a secure environment for sensitive school records.

3. Reliable Backup

By integrating cloud storage (e.g., Google Drive), the database file is protected against data loss due to hardware failure or accidental deletion.

4. *Accessibility and Portability

Storing the database file in the cloud allows authorized users to access or restore it from other locations or devices.

5. Moderate Cost and Complexity

This solution balances functionality with affordability and does not require advanced technical expertise.

Refinements Made

- *User Interface Improvement:* Simplified data entry forms were designed to reduce staff errors and training

time.

- ***Scheduled Backups:** A weekly automatic backup plan was introduced to ensure data is consistently protected.
- ***Basic Training Plan:** A short tutorial guide for teachers and admin staff was created to improve usability.
- ***File Naming and Organization:** A standardized naming format was used for saved files to prevent confusion and ensure easy retrieval.

STAGE 5 PRESENTATION OF RESULTS

Testing the Solution

The student database was tested using real data from 20 students at Mtoko Government High School. Tables for Students, Subjects, and Results were created in Microsoft

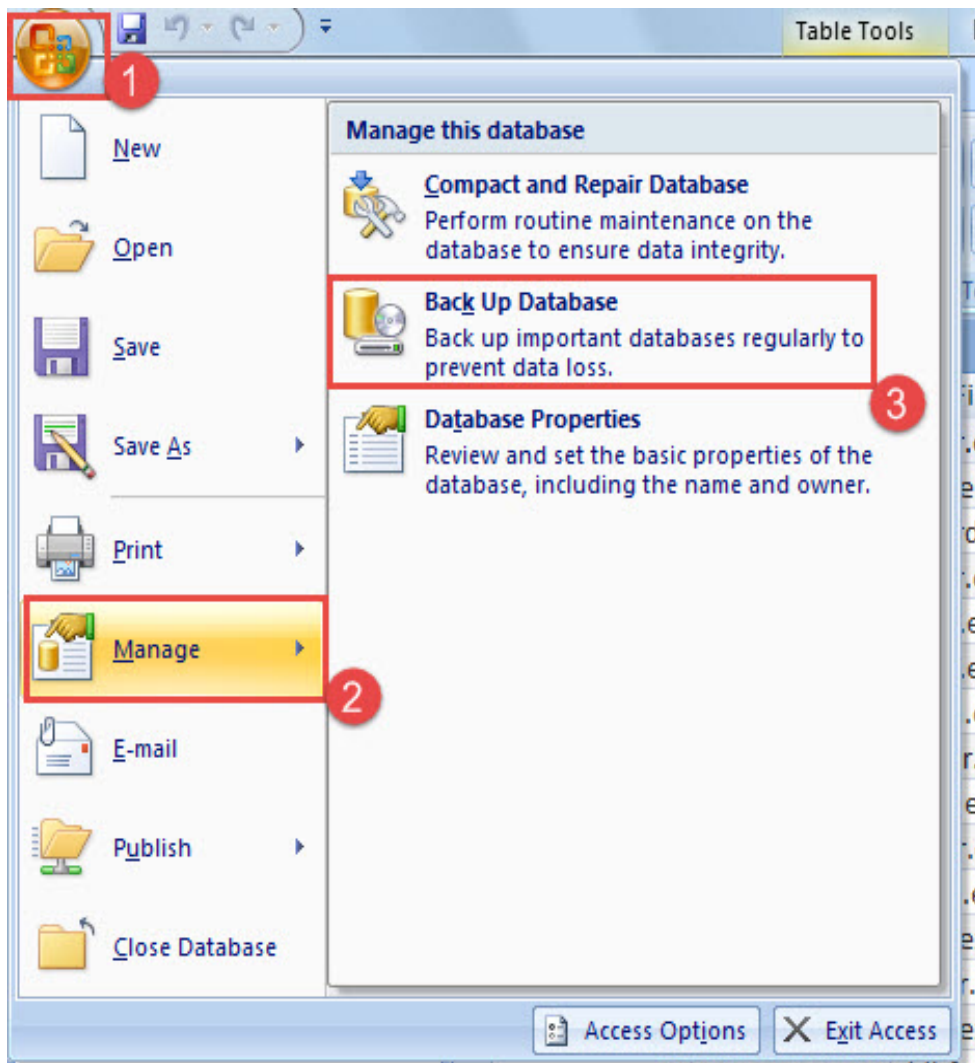
Access. Forms were used for data entry, and queries were run to retrieve student performance and personal information. The database file was also saved and backed up on Google Drive to test cloud storage integration.

Results

- ***Successful Data Entry:*** Student data was entered through user-friendly forms without errors.
- ***Accurate Search Queries:*** Queries retrieved the correct student information based on ID, name, and class.
- ***Generated Reports:*** Academic performance reports were generated and printed using Access.
- ***Secure Backup:*** The database file was successfully uploaded to Google Drive and accessed from another computer.

Effectiveness

- ***Fast and Accurate Information Retrieval:*** Teachers could find student data within seconds.
- ***Data Security and Backup:*** Cloud storage ensured that data remained safe and could be restored if needed.
- [- ***Reduced Paperwork:*** Most records were now digital, minimizing the need for bulky files.
- ***Improved Record Keeping:*** The system minimized errors and ensured updated, consistent records.



STAGE 6 EVALUATION AND RECOMMENDATIONS



Evaluation (Points)

- The database system was easy to use after basic training.
- It significantly reduced time spent searching for student records.
- Reports and queries helped teachers access student performance quickly.
- Cloud backup ensured data security and easy recovery.
- Minor technical skills were needed to manage the system.

Recommendations (Points)

- Train more staff to use Microsoft Access confidently.
- Implement the system school-wide for all student records.
- Schedule regular backups to avoid data loss.
- Upgrade computers to support smooth database operations.
- Consider adding a login system for multiple users with different access levels.

Let me know if you want everything compiled!